

FMDPT-6000 Smart Differential Pressure Transmitter

The FMDPT-6000 is a smart Differential Pressure Transmitter built on monocrystalline silicon piezoresistive sensor technology. It measures the level, density, pressure, and flow of liquid, gas, or steam and converts the pressure signal into a 4-20 mA DC analog current output, with configuration, monitoring, and diagnostics available through HART or Modbus-RTU communication — suited to demanding process and chemical-industry duty.

General Features

- Monocrystalline silicon piezoresistive sensor technology with stable performance, accuracy up to $\pm 0.075\%$ FS.
- Maximum 100:1 turn-down ratio available for flexible measured range adjustment.
- High-performance EMC protection circuit module with strong anti-interference ability.
- Long-term stability: $\pm 0.2\%$ SPAN / 5 years.
- Output via 4-20 mA DC + HART, or Modbus-RTU / RS-485.



Key Specifications

Measurement	Differential Pressure
Range	60 mbar to 100 bar
Accuracy	$\pm 0.075\%$ URL
Turn-Down	100:1
Output	4-20 mA HART, or Modbus-RTU
Stability	$\pm 0.2\%$ Span / 5 years
IP Rating	IP67

Industries & Applications

- Oil and Gas industry
- Food processing
- Pulp and Paper
- Power and Energy
- Chemical Industry
- Marine Equipment

General Specifications — FMDPT-6000

Parameter	Specification
Measurement Type	Differential Pressure
Nominal Range	60 mbar to 100 bar (see range table)
Turn-Down Ratio	100:1
Accuracy (Linear)	±0.075% to ±0.2% URL
Long-Term Stability	±0.2% Span / 5 years
Output Signal	4-20 mA DC + HART, or Modbus-RTU / RS-485
Sensor Technology	Monocrystalline silicon piezoresistive
Power Supply Voltage	HART: 16.5-55 V DC; IS HART: 18.5-28 V DC; Modbus: 12-32 V DC
Power Consumption	4-20 mA: ≤500 mW @ 24 V DC, 20.8 mA; Modbus: ≤240 mW @ 24 V DC, 10 mA
Load Resistance	0-2119 Ohm (operating); 250-600 Ohm (HART communication)
Damping (Electronic)	0-100 s configurable; sensor module damping ≤0.2 s
Certification	Contact Forbes Marshall for current hazardous-area and regulatory approval status
IP Rating	IP67
Weight	About 4 kg (excluding mounting bracket and process connection parts)

Measurement Range Specifications

Nominal Range	Min. Range	Lower (LRL)	Upper (URL)	Line Press.	HP/LP Overload
60 mbar	2 mbar	-60 mbar	60 mbar	250 bar	160 bar
0.4 bar	4 mbar	-0.4 bar	0.4 bar	250 bar	160 bar
2.5 bar	25 mbar	-2.5 bar	2.5 bar	250 bar	160 bar
10 bar	0.1 bar	-10 bar	10 bar	250 bar	160 bar
30 bar	0.3 bar	-30 bar	30 bar	250 bar	160 bar
100 bar	1 bar	-30 bar	100 bar	250 bar	200 bar

Note: Nominal range can also be selected as 20 mbar; consult Forbes Marshall for details. Overload depends on the weakest pressure-bearing component.

Accuracy — Linear Output

Condition	Range	Accuracy
TD ≤ 5	60 mbar (TD ≤ 2 only)	±0.075%
TD ≤ 5	0.4, 2.5, 10, 30, 100 bar	±0.075%
TD > 5	60 mbar, 0.4, 2.5, 10, 30, 100 bar	±(0.001 + 0.0148 × TD)%

Note: Square-root output accuracy is 1.5× linear output accuracy. TD (Turn-Down) = URL / Current SPAN. Reference: 20°C ±5°C, calibrated from zero.

Ambient Temperature Effects

Range	Effect
60 mbar	±(0.1 + 0.05 × TD)% per 10°C of SPAN
0.4, 2.5, 10, 30, 100 bar	±(0.075 + 0.0375 × TD)% per 10°C of SPAN

Environmental Conditions

Parameter	Specification
Working Temperature	-40 to +85°C; With LCD: -20 to +70°C
Storage Temperature	-40 to +100°C; With LCD: -40 to +85°C
Media Temperature	Silicone oil filled: -40 to +105°C; Inert oil filled: -20 to +80°C
Working Humidity	5% RH to 100% RH @ 40°C
Voltage Influence	For 10.5V/16.5V to 55V DC supply, zero/span change ≤0.005% URL/V
Mounting Position Effects	< 4 mbar at any position, correctable by PV = 0 reset
Vibration Effect	< 0.1% URL per GB/T18271.3 / IEC 61298-3

EMC Environment

Test	Standard	Test Condition	Level
Radiated Interference (Casing)	GB/T 9254.1 / CISPR 32	30 MHz - 1000 MHz	Qualified
Conducted Interference (DC Port)	GB/T 9254.1 / CISPR 32	0.15 MHz - 30 MHz	Qualified
Electrostatic Discharge Immunity	GB/T 17626.2 / IEC 61000-4-2	8 kV contact / 8 kV air	A
Radiated RF/EM Field Immunity	GB/T 17626.3 / IEC 61000-4-3	10 V/m (80 MHz - 1 GHz)	A
Power Frequency Magnetic Field	GB/T 17626.8 / IEC 61000-4-8	30 A/m	A
Electrical Fast Transient / Burst	GB/T 17626.4 / IEC 61000-4-4	4 kV (5/50 ns, 100 kHz)	A
Surge Immunity	GB/T 17626.5 / IEC 61000-4-5	1 kV (L-L); 2 kV (L-G)	A
RF Conducted Disturbance Immunity	GB/T 17626.6 / IEC 61000-4-6	3 V/m (150 kHz - 80 MHz)	A

Note: Level A: performance is normal within the limits of the technical specifications.

Power Supply, Load & Time Index

Parameter	Specification
Power Supply Voltage	HART: 16.5-55 V DC IS HART: 18.5-28 V DC Modbus-RTU/RS-485: 12-32 V DC
Load Resistance	0-2119 Ohm (working state); 250-600 Ohm (HART communication)
Transmission Distance	< 1000 m
Power Consumption	4-20 mA: ≤500 mW @ 24 V DC, 20.8 mA Modbus: ≤240 mW @ 24 V DC, 10 mA
Damping (Electronic)	0-100 s configurable
Sensor Module Damping	≤ 0.2 s
Turn-On Time	≤ 6 s
Factory Reset Time	≤ 31 s

Sensor & Wetted Parts Options

Parameter	Code	Description
Diaphragm Material	S / H	S=SS316L; H=Hastelloy C
Fill Oil	S / D	S=Silicone oil; D=Inert filler
Sensor Seal	N / P	N=Low-temp fluorosilicone rubber (-40 to +85°C); P=PTFE square washer (-20 to +70°C)

Electrical Connection (Cable Outlet Protection)

Code	Description	Material	IP Rating
R1	Waterproof connector M20x1.5 + gland, opposite side	PVC	IP67 (6-8mm cable)
R2	Flameproof, 1/2 NPT (F) + gland, opposite side	Stainless steel	IP67 (6-8mm cable)
R3	Flameproof, M20x1.5 (F) + gland, opposite side	Stainless steel	IP67 (6-8mm cable)
R7	Flameproof, G1/2 (F) + gland, opposite side	Stainless steel	IP67 (6-8mm cable)

Output Signal & Display

Code	Description
H	4-20 mA DC + HART, two-wire, power supply 16.5-55 V DC
B	4-20 mA DC + HART, two-wire, intrinsically safe, power supply 18.5-28 V DC
R	Modbus-RTU / RS-485, applicable power supply 12-32 V DC
Display A / C	A = without LCD display; C = with LCD display

Process Connection Options

Code	Specification	Drain Position	Material	Mounting Thread
H1	1/4-18 NPT (F)	Rear end of flange	SS 316	M10x1.5
H2	1/4-18 NPT (F)	Upper part of flange	SS 316	M10x1.5
H3	1/4-18 NPT (F)	Lower part of flange	SS 316	M10x1.5
H4	1/4-18 NPT (F)	Rear end of flange	SS 304	M10x1.5
H7	1/4-18 NPT (F)	Rear end of flange	SS 316	7/16-20 UNF

Ordering Information — Model Code Structure

The FMDPT-6000 model code is structured as follows. Please specify all parameters when placing an order. Factory calibration range (LRL-URL) and display unit must be stated in the purchase contract.

Block	Field	Key Options
Model	Product	FMDPT-6000-DP

Block	Field	Key Options
Sensor	Range	S202D=20mbar; S602D=60mbar; S403D=0.4bar; S254D=2.5bar; S105D=10bar; S305D=30bar; S106D=100bar
Sensor	Diaphragm / Fill Oil / Seal	S/H diaphragm; S/D fill oil; N/P seal
Electrical Connection	Cable Outlet	R1=Waterproof M20; R2=FP 1/2NPT; R3=FP M20; R7=FP G1/2
Output	Signal / Display	H=HART; B=IS HART; R=Modbus-RTU; A=No LCD; C=LCD
Process Connection	Specification	H1/H2/H3/H4/H7 (see table above)
/Options	Fittings	/A1,/A2,/A6,/A7=Process connection adapters
/Options	Mounting	/B1,/B2,/B3,/B6,/B10,/B8,/B11,/B12,/B13=Bracket options (pipe/plate/tube, carbon steel or SS304)
/Options	Other	/Q1=Calibration report; /G1=Ungrease treatment

Code format: the Sensor block (Range + Diaphragm + Fill Oil + Seal) is written as one unbroken code, and the Output block (Signal + Display) is written as one unbroken code; a single dash separates each major block.

Example: FMDPT-6000-DP-S105DSSN-R2-HC-H1/B6 [CAL: 0 to 10 bar]

Dimensional Reference (mm)

Parameter	With Display	Without Display
Overall Width	131 mm (93 mm body)	131 mm (93 mm body)
Overall Depth	166 mm	166 mm
Overall Height	210 mm	210 mm
Base Width	90 mm	90 mm
Mounting Bolt Spacing	54 / 80 mm	54 / 80 mm

Note: Product is factory-assembled with NPT plug and drain valve. Adapter options (/A1, /A2, /A6, /A7) and mounting brackets change overall envelope; consult Forbes Marshall for adapter-specific drawings.

Quick Reference — Common Configurations

Worked examples for reference only; confirm every field against the full model code table before ordering.

Configuration	Order Code	Range
Standard DP, SS316L, silicone oil, HART, LCD	FMDPT-6000-DP-S105DSSN-R1-HC-H1	0-10 bar
Hastelloy diaphragm, Modbus, no LCD, flameproof	FMDPT-6000-DP-S105DHDN-R2-RA-H1	0-10 bar